INVENTORY AND PROCUREMENT DECISION-SUPPORT SYSTEM

Using Demand Forecasting and Lead-Time Planning

CSIS 4495 – Applied Research
Section 002

Mid-Course Progress Report

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Mid-Course Demo Video



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1. Submission Checklist and Introduction

Submission Checklist

Done	Requirement
☐	GitHub repository contains all work completed so far
☐	One joint video demo is at least 6 minutes
☐	Both team members present/demo their own work in the video
☐	Public clickable video link is placed on the title page
☐	README contains project description and installation instructions
☐	Completed report is submitted on Blackboard by the team lead
☐	Completed report is also checked into GitHub
☐	Both students attend the mandatory in-person check-in

Introduction

Inventory and procurement planning is hard when a company has many products, many suppliers, long lead times, and shipments that arrive on different dates. Knowing the current stock is not enough to decide when to place the next order. A buyer also needs to look at past sales, orders that are on the way, the supplier's lead time, and possible delays. Lead time means the number of days between placing an order and receiving it.

This project comes from a problem that is common in everyday purchasing work. The information is often spread across several spreadsheets or files, so the user has to check many places before making a decision. Our project brings the main information together in one web application. It gives a high-level view for managers and a detailed view for each product. In this report, a SKU (stock keeping unit) means one product item that is tracked in inventory.

We are building the system in Python with Streamlit, pandas and Plotly. At this stage, the application can show overall business information, detailed information for each SKU, a basic demand forecast, a projection of future inventory, shortage risk, and a basic reorder recommendation. We are also adding supplier performance and scenario testing.

Research Questions

1. Can historical sales provide a useful baseline demand forecast?
2. How do current stock, incoming purchase orders and lead time affect future inventory?
3. Can the system identify a shortage early enough to support a reorder decision?
4. How do higher demand and shipment delays change the result?
5. How can the same data be shown clearly to both management and procurement users?

2. Initial Project and Changes to the Proposal

Initially Proposed Project

Our original project looked at inventory planning for one product at a time. The user would pick a SKU and see its current stock, past sales, incoming purchase orders, supplier lead time, a demand forecast, projected inventory, the month a shortage may happen, and a suggested reorder date and quantity.

The original problem was based on products that can take three to four months to replenish. One SKU can also have several incoming shipments that arrive on different dates. For this reason, a decision cannot be made by looking only at the current inventory.

Main Changes Since the Proposal

Change	Reason
Executive dashboard	The first working version was too detailed and started directly at SKU level. Based on instructor feedback, we added a bigger-picture management view first.
Expanded dataset	The original synthetic dataset was much smaller. It was expanded to about 300 SKUs with a longer sales history, selling prices, revenue and richer purchase-order data.
Financial view	Revenue, estimated gross profit, inventory value and open PO value were added so the dashboard is useful to a manager, not only a buyer.
Supplier performance	Supplier reliability is now part of the planned system because supplier lead time and delays directly affect replenishment.
Scenario testing	What-if controls for higher demand and shipment delays are being added so the system can show how a recommendation changes under different conditions.

Reason for the Changes

We made these changes mainly to turn the project into a complete decision-support system that is easy to understand. A manager should be able to open the application and quickly see how the business is doing, what needs attention, and where to look next. The detailed SKU page is still important, but it now works as a drill-down page instead of being the only view.

3. Updated Planning, Timeline and Responsibilities

Team Responsibilities

Team Member	Main Responsibilities
Anushka	Python/Streamlit setup, application development, dashboard, charts, forecast, inventory projection, reorder logic, scenario testing, technical testing and documentation.
Jasleen	Dataset review, product/inventory/PO understanding, procurement business rules, validation of purchasing logic, testing, evaluation notes and project documentation.
Both	Project planning, feature decisions, GitHub updates, report, demo video and final testing.

Updated Timeline - Mid-Course to Final Submission

Period	Main Work	Status
Oct 19-25	Executive dashboard, larger dataset, financial KPIs, supplier/scenario development	In progress / mostly completed
Oct 26-28	Mid-course report, demo video, README, GitHub check-in, in-person review	Mid-course submission
Late Oct - Early Nov	Test more SKUs, improve scenario planner and supplier analysis	Planned
November	Forecast evaluation, compare baseline with another method, improve reorder logic	Planned
Late November	SQLite integration if useful, UI cleanup, full test cases	Planned
Final weeks	Final evaluation, report, demo and repository cleanup	Planned

Simple Gantt-Style Plan

Work Area	Oct 19-25	Oct 26-28	Early Nov	Mid Nov	Final Weeks
Dashboard & SKU analysis	■	■			
Scenario & supplier work	■	■	■		
Forecast evaluation			■	■	
Reorder logic testing			■	■	
UI / database cleanup				■	■
Final report & demo					■

■ = *planned work period. This timeline may change slightly after the mid-course feedback.*

4. Implemented Feature - Executive Dashboard

One of the biggest changes in this period was adding an executive-level dashboard. The earlier version of the app focused on one SKU at a time. That was useful for detailed analysis, but it did not give a manager a quick picture of the whole company.

The dashboard is made for a user who may have only a few minutes to check the business. It shows year-to-date sales, revenue, inventory value, open purchase order value, top-selling products, delayed purchase orders, and products that may need attention. Figures 1 and 2 show the dashboard.

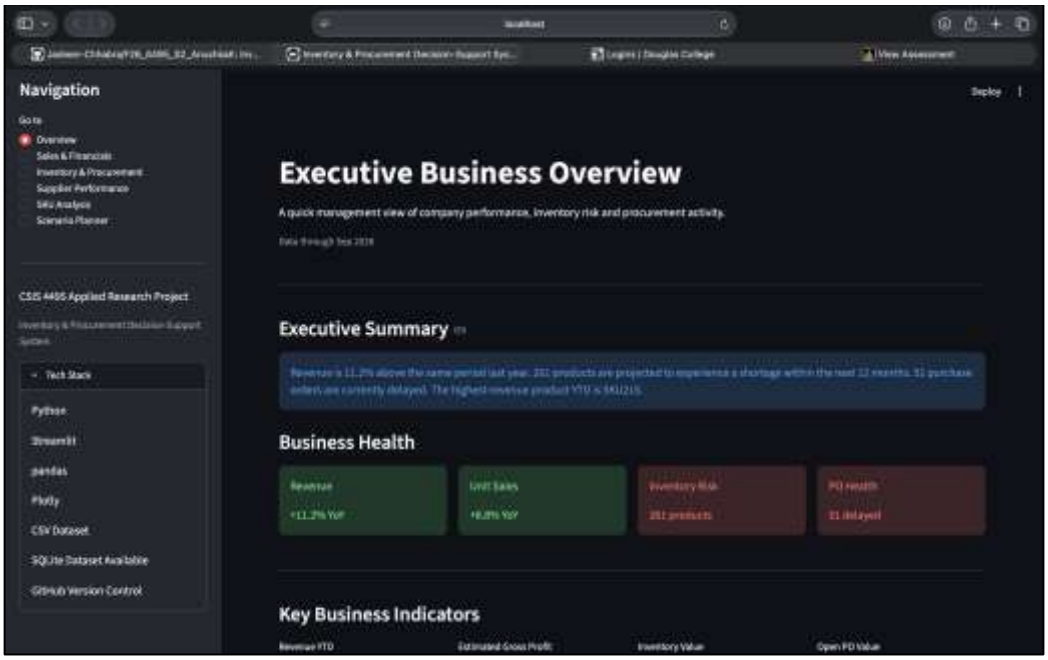


Figure 1: Executive Business Overview page of the application

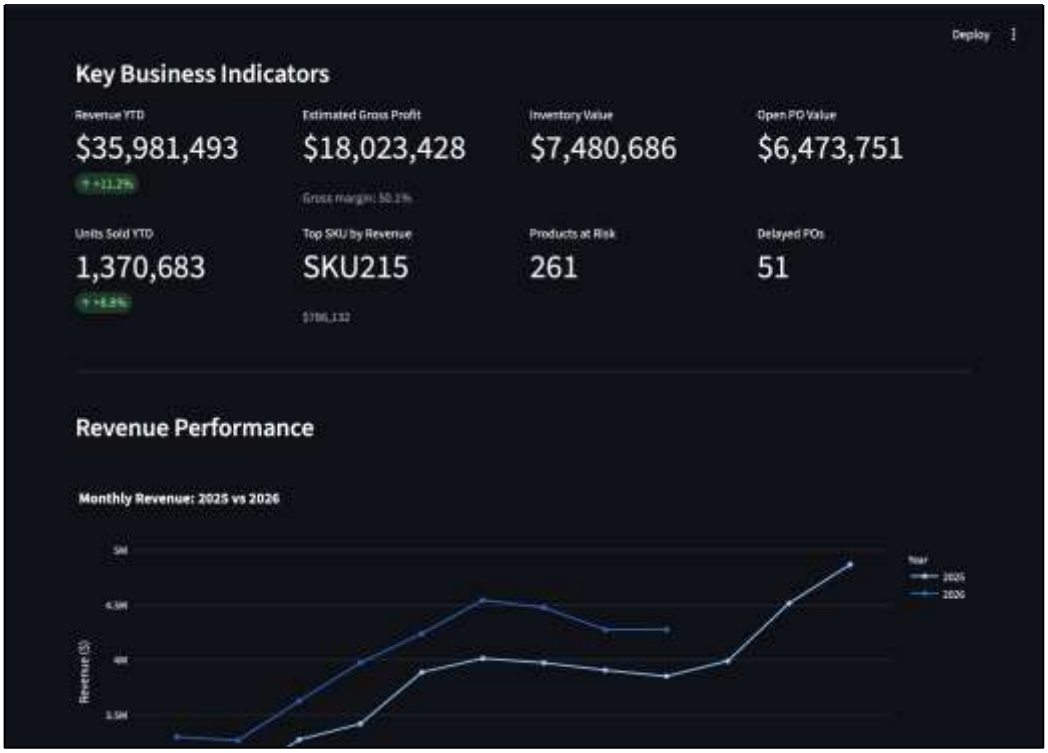


Figure 2: Key business indicators and monthly revenue comparison (2025 vs. 2026)

The dashboard also compares the current year with the previous year, so it is easy to see whether sales and revenue are going up or down. The goal of the first page is not to show every detail. It answers three simple questions: How is the business doing? What needs attention? Where should the user look next?

Management-Level Indicators

- Revenue year to date and year-over-year change
- Estimated gross profit and margin
- Inventory value and current inventory units
- Open PO value and delayed PO count
- Top SKU by revenue and by units
- Number of products projected to run short

5. Implemented Feature - SKU Analysis and Demand Forecast

The SKU Analysis page is the detailed part of the application. The user selects a product, and the app combines data from the product master, inventory, purchase orders and monthly sales. This keeps the original purpose of the project while the executive dashboard stays simple. Figure 3 shows the page.

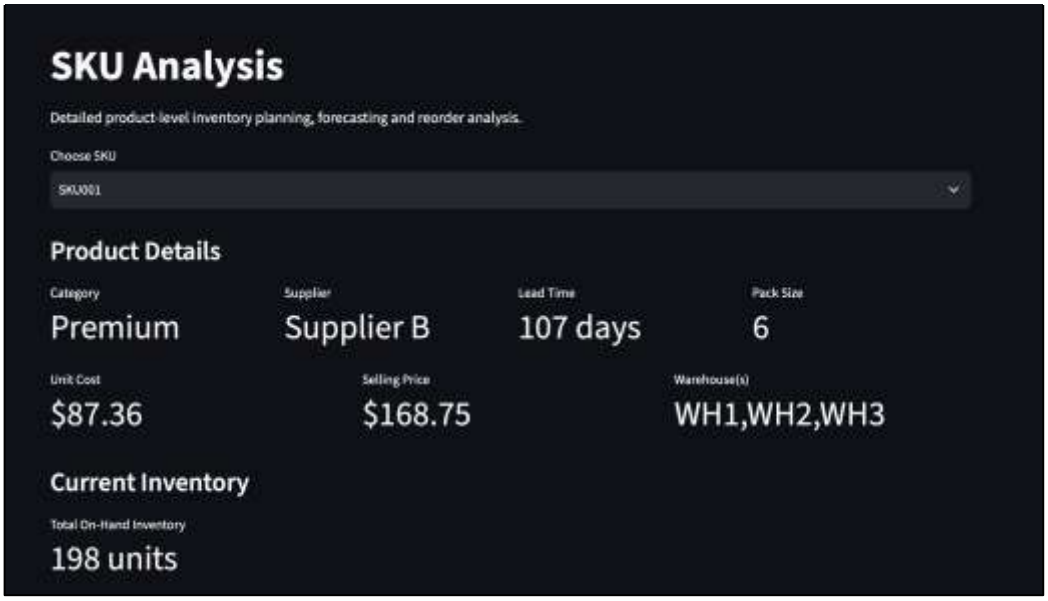


Figure 3: SKU Analysis page showing product details for SKU001

For the selected SKU, the application shows the category, supplier, lead time, pack size, unit cost, selling price, warehouses, current stock and open purchase orders (Figures 3 and 4). The monthly sales history is shown next, so the user can see the demand pattern before looking at the forecast (Figure 5).

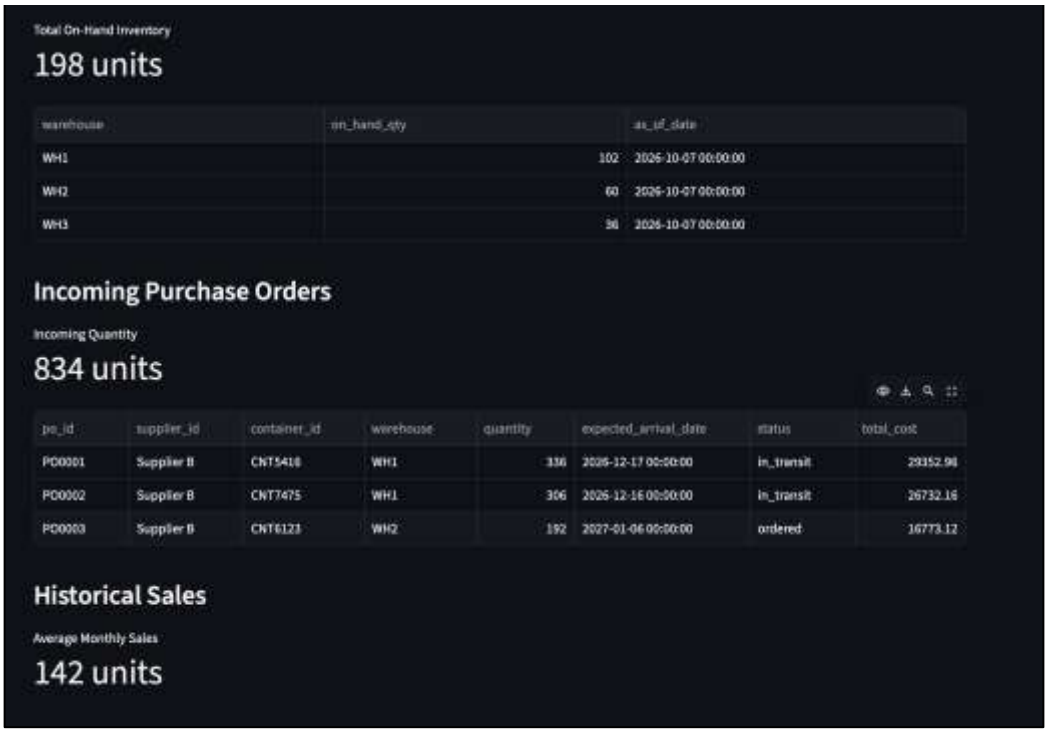


Figure 4: Current inventory by warehouse and incoming purchase orders for SKU001

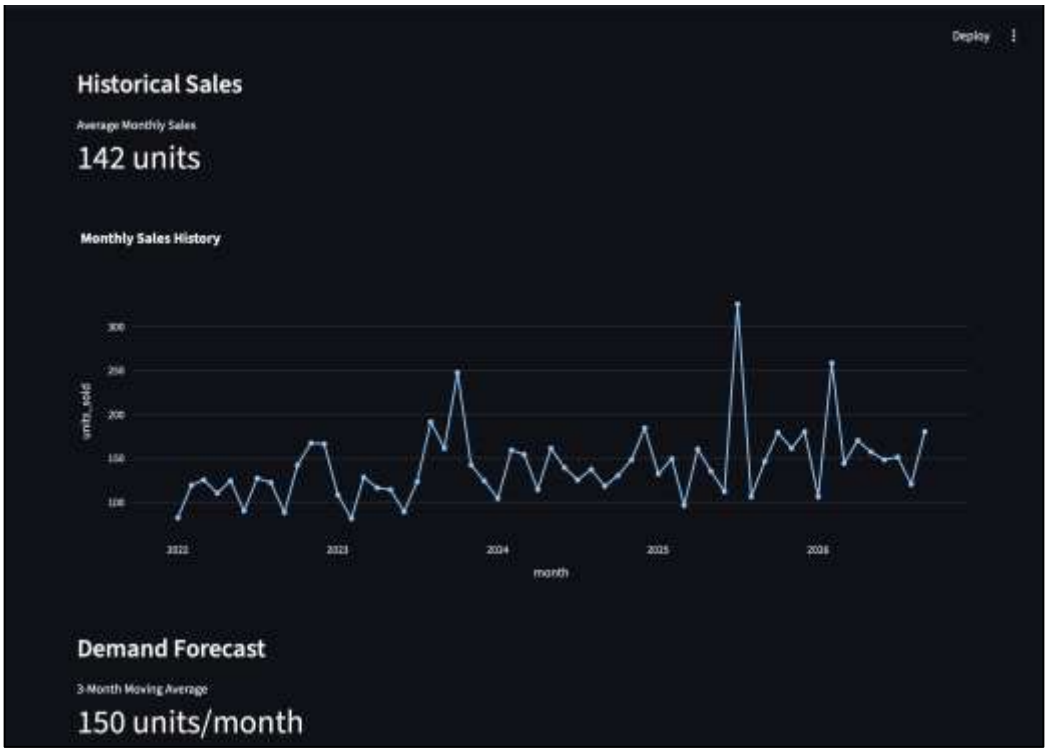


Figure 5: Monthly sales history and three-month moving average forecast for SKU001

Baseline Forecast

The first forecasting method is a three-month moving average. The system takes the sales of the last three months and calculates their average. We chose it as a baseline because it is simple to understand and test, and it gives us a starting point to compare with another method later.

The moving average is not our final forecasting model. In the second half of the project, we plan to test how accurate it is using historical data and compare it with another method.

6. Inventory Projection and Reorder Recommendation

After the demand forecast is calculated, the application builds a 12-month inventory projection. For each month, it starts with the opening inventory, adds the purchase orders expected that month, and subtracts the forecast demand.

The calculation is: Closing Inventory = Opening Inventory + Incoming Purchase Orders − Forecast Demand. The closing balance of one month becomes the opening balance of the next month (Figures 6 and 7).

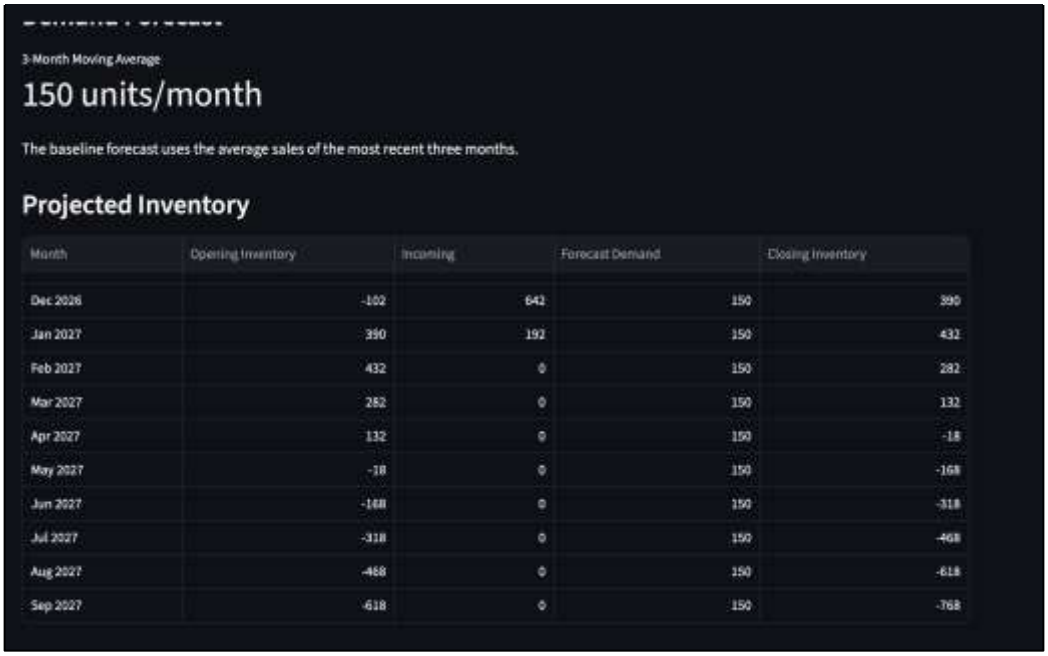


Figure 6: Twelve-month projected inventory table for SKU001



Figure 7: Projected inventory chart showing the month when stock becomes negative

If the projected closing inventory becomes negative, the application finds the first month where a shortage is expected. This gives the user an early warning before the current stock is actually low.

Basic Reorder Recommendation

When a shortage is found, the system works backwards from the shortage month using the supplier's lead time. This gives a suggested order date. The first rule for the reorder quantity is about three months of forecast demand, rounded to the product's pack size (Figure 8).



Figure 8: Reorder recommendation with suggested order date and quantity

This is a basic rule on purpose at the mid-course stage. Our goal was to complete one full flow from sales history to forecast, inventory projection and purchase recommendation. We will test and improve the reorder logic later.

7. Scenario Testing and Supplier Performance

Scenario Testing

We are adding scenario testing so the user can see what happens when the original plan changes. The user can increase demand by 10%, 20% or 30%, and can delay incoming shipments by 15, 30 or 45 days. The system then recalculates the inventory projection, the shortage month and the reorder recommendation (Figures 9 and 10).

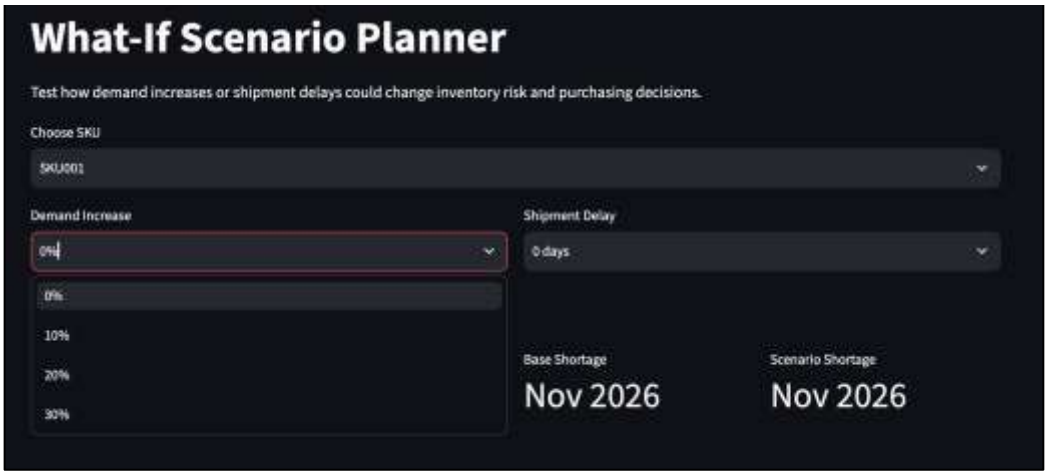


Figure 9: What-If Scenario Planner with demand increase options



Figure 10: Base versus scenario inventory projection with shipment delay options

This feature matters because real purchasing does not always follow the forecast. A sudden rise in sales or a late container can change the date when stock runs out.

Supplier Performance

We are also adding supplier performance because a supplier's reliability directly affects the risk of running short. The expanded purchase order data includes the supplier, order date, expected arrival date, actual arrival date and order status. With this, the project can calculate useful supplier measures instead of only showing the supplier name (Figures 11 to 13).

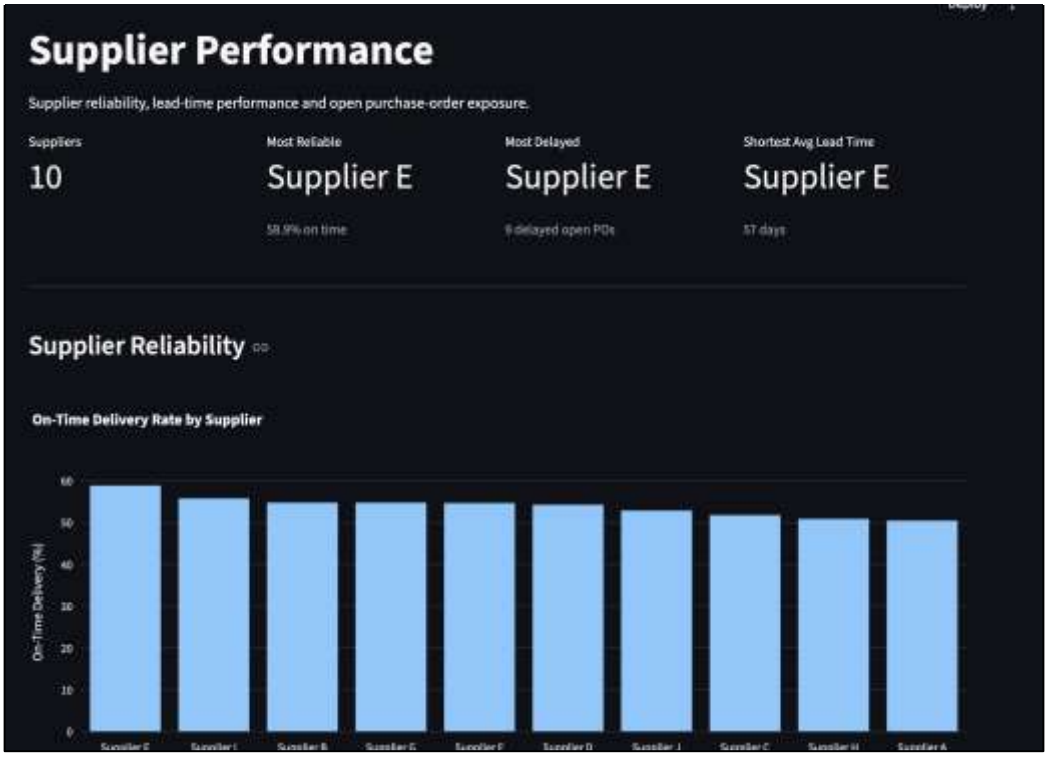


Figure 11: Supplier Performance page with reliability indicators

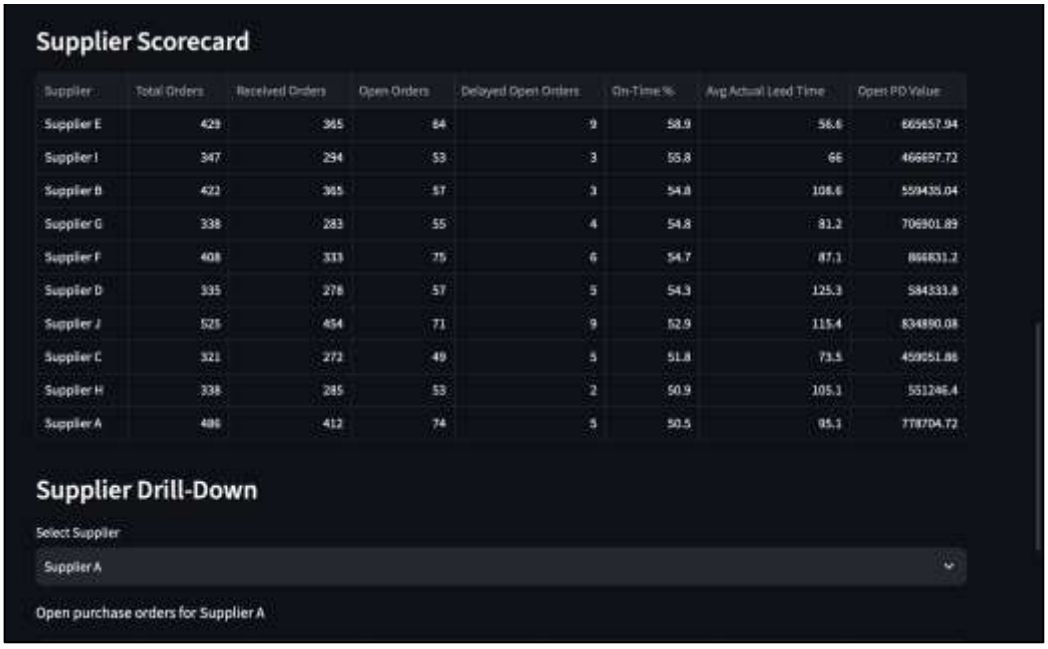


Figure 12: Supplier scorecard table with on-time delivery and lead time

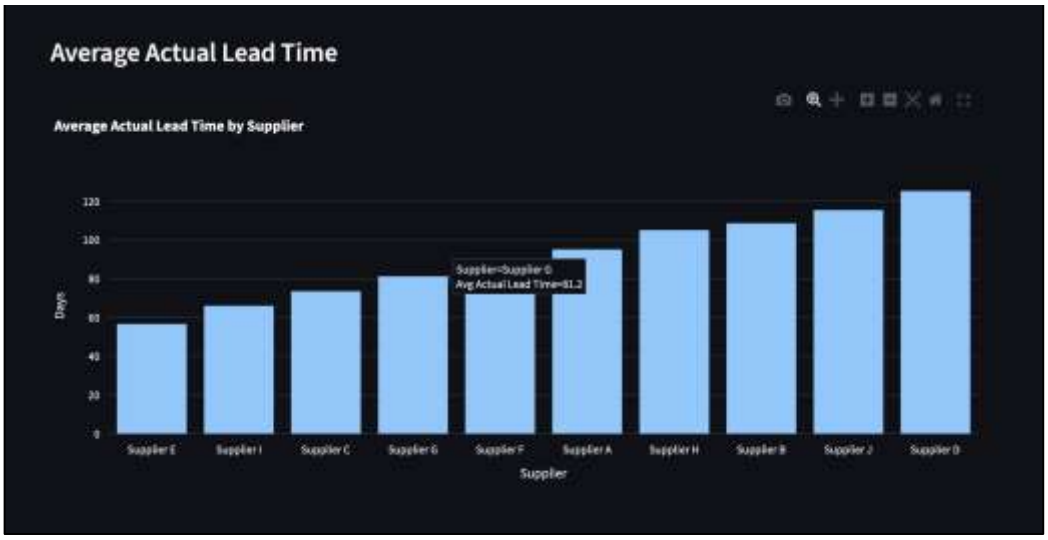


Figure 13: Average actual lead time by supplier

The supplier measures include average actual lead time, on-time delivery percentage, delayed open orders, number of received orders and open purchase order value. These help the user understand why a product may be at risk.

8. Technical Implementation, Testing and Issues

Technology Used

Technology	Use in Project
Python 3.11	Application logic and calculations
Streamlit	Web interface and navigation
pandas	Loading, filtering, grouping and joining data
Plotly	Interactive charts
CSV	Current working data source
SQLite	Generated dataset version available for later use
GitHub	Version control and progress tracking
VS Code	Main development environment

Main Issues Encountered

Issue	What We Did
App was too SKU-focused	Added an executive dashboard and separate navigation so the app starts with the bigger picture.
Original dataset was too small	Expanded to about 300 SKUs with longer history, prices, revenue and richer PO records.
New PO structure caused a code error	The newer PO file already had unit_cost and total_cost. The duplicate merge was removed and total_cost is now used directly.
Forecast/reorder logic could become too complex too early	Kept a simple moving-average baseline and basic reorder rule so one complete feature works end to end before adding more advanced methods.
Need to show individual team progress	Both team members are keeping separate contributions and GitHub/work-log entries for the report and demo.

Current Testing

Our testing so far is mainly functional testing. We check that the app loads the dataset correctly, that different SKUs can be selected, that inventory and purchase orders match the selected product, and that the forecast and projection change when the data changes. We also test different situations, such as normal stock, low stock, incoming orders and delayed orders.

9. Evaluation Plan and Remaining Work

Forecast Evaluation

The project still needs a formal evaluation. We plan to use historical sales where the real values are already known. We will use part of the history to make a forecast and then compare it with the actual sales. We can report the difference with Mean Absolute Error (MAE) or Mean Absolute Percentage Error (MAPE).

Forecast Method Comparison

We will treat the three-month moving average as the baseline and compare it with at least one other forecasting method. Our aim is not to use a complicated model just for the sake of it. We want to see whether another method gives a real improvement for this type of inventory planning problem.

Scenario and Business-Logic Testing

We will check scenario testing using cases where demand goes up or incoming shipments are late. We will confirm that the inventory projection changes in the expected direction and that the shortage and reorder recommendation also change when needed.

Remaining Work After Mid-Course

- Test the application across a larger set of SKUs and edge cases.
- Finish and validate supplier-performance calculations.
- Improve the scenario planner and comparison output.
- Evaluate forecast accuracy using historical hold-out data.
- Compare the moving-average baseline with another forecasting method.
- Review and improve the reorder quantity/date logic.
- Consider using SQLite directly instead of CSV files if it improves the final implementation.
- Clean up the interface and complete final documentation, testing and demo.

In the second half of the project, we will focus on evaluation and testing instead of adding many unrelated features. Our goal is to show that the system gives useful decision support that is easy to understand.

10. AI Use and Individual Work Logs

AI Use

Generative AI was used as a support tool in this project. It was used mainly to explain errors, discuss possible application structures, help us understand Python and Streamlit code, solve problems, and organize documentation. We reviewed and tested the output instead of assuming that AI-generated code or text was correct.

AI Tool	Version / Account	Specific Use	Value Added by Us
ChatGPT (OpenAI)	Plus account; confirm model/version used	Explaining Python/Streamlit errors, discussing dashboard structure, code troubleshooting, documentation support	We selected the business requirements, tested the code, changed features based on instructor feedback, validated results and decided what was useful for procurement.

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Work Log - Anushka

Date	Hours	Work Completed
Oct 7, 2026	1.5	Met with Jasleen and reviewed public datasets for the project.
Oct 11, 2026	1.0	Met with Jasleen and finalized the project data approach.
Oct 13–14, 2026	2.0	Set up Streamlit and built the first SKU-level application with inventory, PO and sales views.
Oct 14–15, 2026	1.5	Added moving-average forecast, 12-month inventory projection, shortage detection and reorder recommendation.
Oct 19–21, 2026	2.5	Built and improved the Executive Dashboard with management KPIs and business-health indicators.
Oct 24–25, 2026	1.5	Updated the app for the expanded 300-SKU dataset and fixed the PO cost issue.
Oct 25–26, 2026	1.5	Added financial KPIs, Supplier Performance and Scenario Planner features.
Oct 26, 2026	1.0	Tested the application and worked on the Mid-Course Report, README and demo preparation.

Work Log - Jasleen

Date	**Hours**	**Work Completed**
Oct 7, 2026	0.5	Reviewed the project proposal again to list the data our system needs (sales history, stock, purchase orders with ETAs, lead times, pack sizes).
Oct 11, 2026	1.5	Morning: studied public dataset websites (M5 Forecasting, UCI Online Retail II, DataCo Smart Supply Chain).
Oct 13–14, 2026	1.5	Evening: met with Anushka, continued the review, and shortlisted the resources that are valuable for our project.
Oct 14–15, 2026	1.0	Met with Anushka and finalized the data we will use for the project, based on what the public datasets do and do not cover.
Oct 19–21, 2026	2.0	Generated the project dataset (45 SKUs, sales, inventory, purchase orders) with a Python script using AI assistance. Rechecked all the spreadsheets myself.
Oct 24–25, 2026	1.0	Checked the data files (dates, stock and purchase order values) and checked them in to the repo under Implementation/Data.
Oct 25–26, 2026	1.5	Expanded the dataset to 350 SKUs with selling prices, revenue and fictional product and supplier names (AI assisted), and rechecked the new files.

Date	Hours	Work Completed
Oct 26, 2026	1.0	Helped define the procurement rules and expected outputs for the reorder recommendation. Tested SKU situations and reviewed the purchasing logic. Worked on documentation, evaluation notes and report sections.

11. Closing, References and Prompt-History Appendix

Closing

At the mid-course stage, the project has moved from an idea to a working Streamlit application. The system can combine past sales, current inventory and incoming purchase orders for individual products. It can create a basic forecast, project future inventory, find possible shortages and give a first reorder recommendation.

The project has also moved toward a bigger management view. The executive dashboard, financial indicators, supplier information and scenario testing make it a complete decision-support application and not only a product-level calculator.

The next stage will focus on testing and evaluation. We plan to measure forecast accuracy, compare forecasting methods, improve the scenario and supplier analysis, and review the reorder logic. Our overall goal is still to make inventory and procurement information easier to understand and act on.

References

- CSIS 4495 Applied Research - Mid-Course Progress Report Template and course instructions.
- Streamlit documentation - used for application structure and interface development.
- pandas documentation - used for data loading, filtering, grouping and calculations.
- Plotly documentation - used for interactive charts.

Appendix A - Prompt History

AI was used mainly for troubleshooting and explanation. Add the prompt history required by the course here. A short index can stay in the report, and the full exported or chat history can be attached or pasted if the instructor expects every prompt.

Prompt Area	**Purpose**
Project setup	Help set up Python, Streamlit and the project folder/environment.
Dataset selection	Compare possible datasets and decide what fields are needed for inventory/procurement.
SKU page	Create a page showing product, stock, purchase orders and monthly sales.
Forecast	Explain and implement a simple three-month moving-average forecast.
Projection	Project inventory using opening stock, incoming POs and forecast demand.
Dashboard	Create a bigger-picture executive dashboard after instructor feedback.
Expanded data	Update the application for the 300-SKU synthetic dataset and new PO structure.
Scenario / supplier	Discuss scenario testing and supplier-performance analysis for the final system.